AKASH JADON

LEAD AI SYSTEMS ARCHITECT & FOUNDING FULL-STACK ENGINEER

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PROFESSIONAL SUMMARY

Senior AI Systems Architect and High-Agency Full-Stack Engineer with 3+ years engineering autonomous LLM orchestration kernels, distributed concurrency pipelines, and production SaaS architectures. Architect of Astra AI Core (4,400+ LOC custom Python streaming runtime with per-provider FSM circuit breakers and live token interceptors), BossConsole (native JVM/Kotlin desktop harness executing 100+ MCP tools with sub-110MB RAM footprint), and DevisBat (commercial Swiss trade fintech platform with Gemini Vision estimation and QR-bill generation). Deep expertise in low-latency token streaming (sub-120ms TTFT), atomic Redis locking (SETNX), and production-hardened web applications on Next.js 15, TypeScript, and AWS.

TECHNICAL ARSENAL

Agentic AI & Orchestration:	Autonomous ReAct execution loops, Model Context Protocol (MCP), streaming <think> token state machines, per-provider FSM circuit breakers, RAG, Swappable LLMs (Claude 3.5 Sonnet, Gemini 2.5, DeepSeek-V3, Groq).
Concurrency & Reliability:	Distributed Redis atomic locks (SETNX), SHA-256 canonical idempotency hashing, token-bucket rate limiting, exponential backoff retries with jitter, Dead-Letter Queues (DLQ), zero-hallucination pipelines.
Full-Stack & Cloud Infra:	Next.js 15 (App Router, Server Actions), React 19, TypeScript, Tailwind CSS, Python (FastAPI, asyncio, Uvicorn), PostgreSQL (Row-Level Security), Supabase, Redis, Docker, AWS (ap-south-1), Vercel Edge Runtime.
Security & Protocols:	AES-256-GCM record-level encryption, Webhook HMAC validation, Meta WhatsApp Cloud API, Telegram Business API, Meta Conversions API (CAPI), Close CRM API, Capacitor Android SDK native compilation.

REPRESENTATIVE PRODUCTION ARCHITECTURES

Astra AI Core — Autonomous AI Runtime & Streaming Kernel

Python / FastAPI / Playwright / SSE / Supabase

- Kernel Architecture & Sub-120ms TTFT: Engineered a 4,400+ LOC custom Python streaming backend utilizing Server-Sent Events (SSE) and async connection pooling, achieving sub-120ms Time-To-First-Token across 4 model providers.
- FSM Circuit Breaker Failover: Built Finite State Machine circuit breakers monitoring real-time error rates across NVIDIA GLM, DeepSeek, Gemini, and Groq; slashed streaming dropout rates from 14.2% to <0.05% with zero user interruption.
- Streaming <think> Token Interceptor: Built real-time token state machine intercepting reasoning tags on live streams, stripping client clutter, rendering collapsible UI accordions, and triggering automated Playwright web research upon uncertainty detection.
- Autonomous Sandboxed ReAct: Deployed self-healing browser automation loops inside headless Playwright sandboxes backed by dual-key rotation vaults and 500-slot semantic LRU response caching.

BossConsole — Native JVM Multi-Threaded AI Agent Operator Console

Kotlin / Compose Desktop / JVM 21 / MCP / Open Source

- Native Desktop Runtime (110MB RAM): Architected an open-source JVM/Kotlin Compose Desktop agent operator console, cutting resident memory from 850MB+ (typical Electron wrappers) down to <110MB while orchestrating Claude Code, Gemini, and Codex.
- 100+ MCP Tool Harness: Implemented hot-reloadable Model Context Protocol (MCP) server/client harness enabling agents to dynamically adapt and discover tool interfaces at runtime without restarting.
- Encrypted QR Mobile Handoff: Built end-to-end encrypted tunneling bridge pairing via dynamic QR code, allowing operators to monitor agent CLI terminal execution in real time from mobile devices. Packaged for macOS DMG, Windows MSI, Linux.

DevisBat (swissquote-ai) — Commercial Swiss Trade Quoting & Fintech SaaS

Next.js 15 / Supabase RLS / Gemini Vision / Swiss QR-Bill

- Production Swiss B2B SaaS: Built commercial platform converting technician voice notes and site photos into priced commercial quotes and legally compliant Swiss QR-bill invoices across 12 trade categories, cutting quote turnaround from 4 hours to 90 seconds.
- 16,800+ SKU Matching Engine: Engineered catalogue matching engine against a 16.8k item database with automated Gemini Vision estimation fallback, Swiss canton wage tables, travel surcharges, and Supabase Row-Level Security.
- Fintech & Statutory Compliance: Integrated Swiss canton-specific labour wage tables, travel surcharges, material margin rules, Swiss VAT (8.1%), and automated QR-bill invoice generation with multi-tenant company isolation and trial lifecycle state machines.

AstraQuote Lead Engine — Concurrency Pipeline & Verification Engine

Python / Redis SETNX / DuckDuckGo / E.164 / Open Source

- Atomic Concurrency Control: Architected multi-source ingestion pipeline with canonical SHA-256 fingerprinting and atomic Redis locks (SETNX), eliminating 100% of race conditions and duplicate API spend across 50,000+ operations.
- Deterministic Verification: Enforced zero-hallucination pipelines via automated E.164 phone formatting, Swiss canton validation, and Dead-Letter Queues (DLQ) with automated Slack alerting. Code audited and open-source on GitHub.
- Resilience & Rate-Limiting: Deployed token-bucket rate limiters with exponential backoff and jitter across multi-source discovery scrapers, maintaining 99.98% extraction reliability under aggressive target anti-bot throttling.

Evolnex Relay — AES-256 Encrypted Omnichannel Edge Gateway

Next.js Edge Runtime / Upstash Redis / WhatsApp Cloud API

- Zero-Downtime Edge Routing: Built sub-80ms edge messaging gateway integrating WhatsApp Cloud API, Telegram Business API, and Gmail OAuth on Vercel Edge Runtime with record-level AES-256-GCM encryption and shadow-mode AI triage.
- Enterprise Security Architecture: Built record-level AES-256-GCM encryption with cloud-separated keys, webhook HMAC signature validation, intelligent shadow-mode AI triage, and fail-safe human takeover controls.
- Distributed Edge Streaming: Processed high-volume webhook bursts using Upstash Redis streaming queues on Vercel Edge Runtime, guaranteeing sub-80ms p95 delivery to avoid provider retry storms.

PROFESSIONAL EXPERIENCE

Founder & Lead Systems Architect — Evolnex Technologies

2023 – Present | Remote

- Direct technical architecture and engineering delivery for an agile 7-engineer software studio delivering bespoke enterprise AI platforms, high-throughput automation pipelines, and mission-critical web applications.
- Architected and shipped mission-critical web platforms, Progressive Web Apps (PWAs), and institutional proposals (including Aaranyak Conservation HRMS and West Bengal Primary LMS).
- Led development of proprietary SaaS products (DevisBat, Astra AI Core, Relay), managing full software development lifecycles from initial Figma UX design tokens to production Kubernetes/AWS cloud deployments.
- Established engineering standards across the team: engineered CI/CD pipelines, Dockerized runtime harnesses, deterministic LLM evaluation suites, and strict async/await concurrency patterns.

KEY INSTITUTIONAL & CLIENT PLATFORMS

- Aaranyak Conservation HRMS: Architected 10-module enterprise HRMS & field appraisal platform for UNESCO World Heritage zones, featuring offline-first biometric authentication, GPS geofencing, and automated payroll pipelines.
- West Bengal Primary LMS Platform: Designed dual-portal learning management platform (11,000+ teachers, 100k+ students) with sub-100ms assessment evaluation engines and EduRev-compliant real-time data sync.
- Fountain Life HealthTech: Engineered high-conversion marketing platform for premier longevity membership brand (Longevity Clinic of the Year 2025) reaching high-net-worth members globally.
- Girl Effect (WAZZII Bot): Architected multi-lingual WhatsApp conversational bot and content moderation filters for adolescent reproductive health across East Africa, ensuring zero-hallucination statutory safety.

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

Certificate in Automation and Agentic AI

Udemy | 2026

Advanced curriculum covering autonomous agentic execution loops, multi-agent orchestration, tool calling, and enterprise workflow design.

Certificate in Full-Stack Software Engineering

Udemy | 2026

Comprehensive engineering credential covering Next.js 15, React 19, TypeScript, Python FastAPI, PostgreSQL RLS, REST/GraphQL APIs, and cloud deployments.

Registered Micro Enterprise (MSME Udyam)

Govt. of India | Verified

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